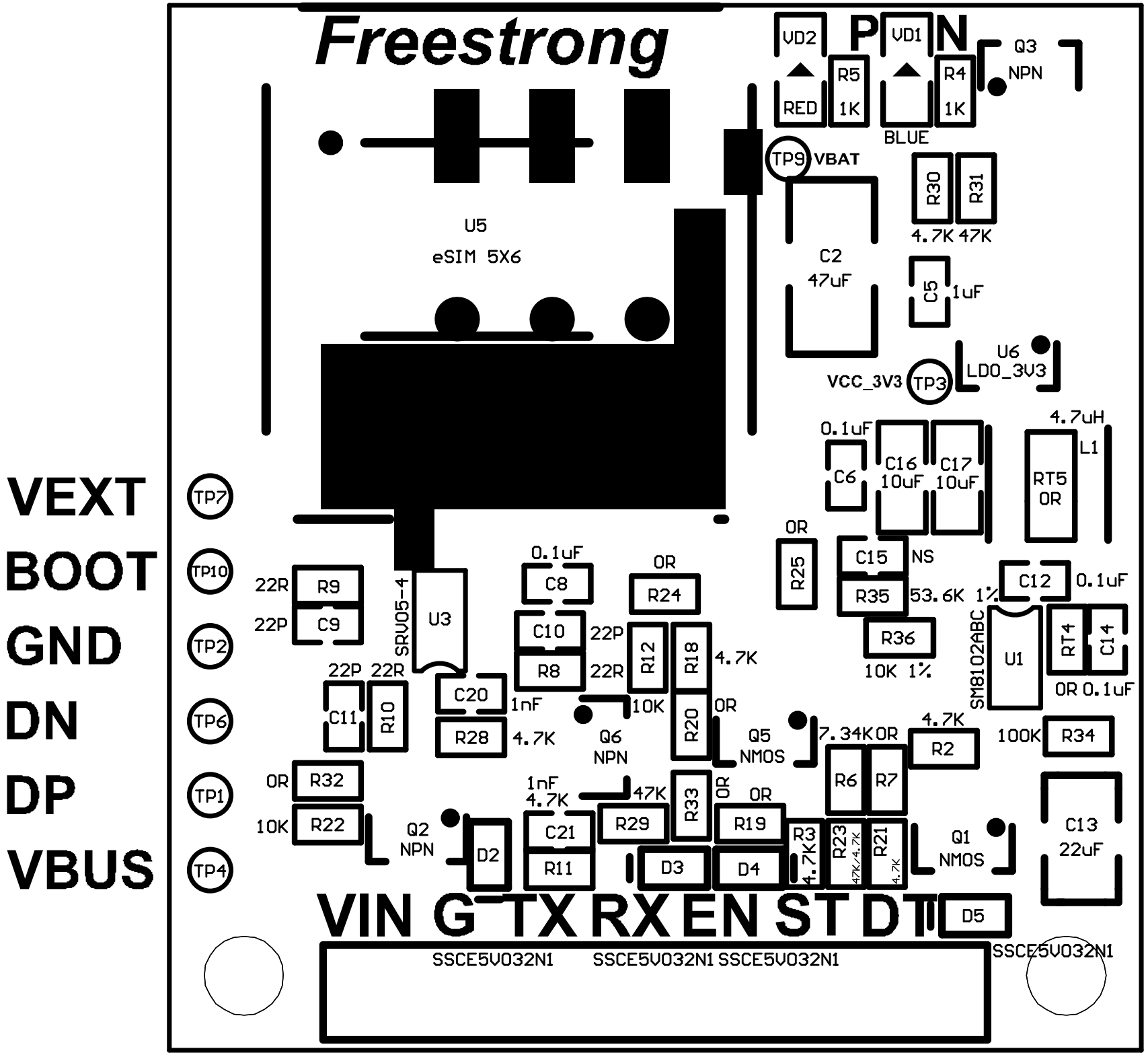
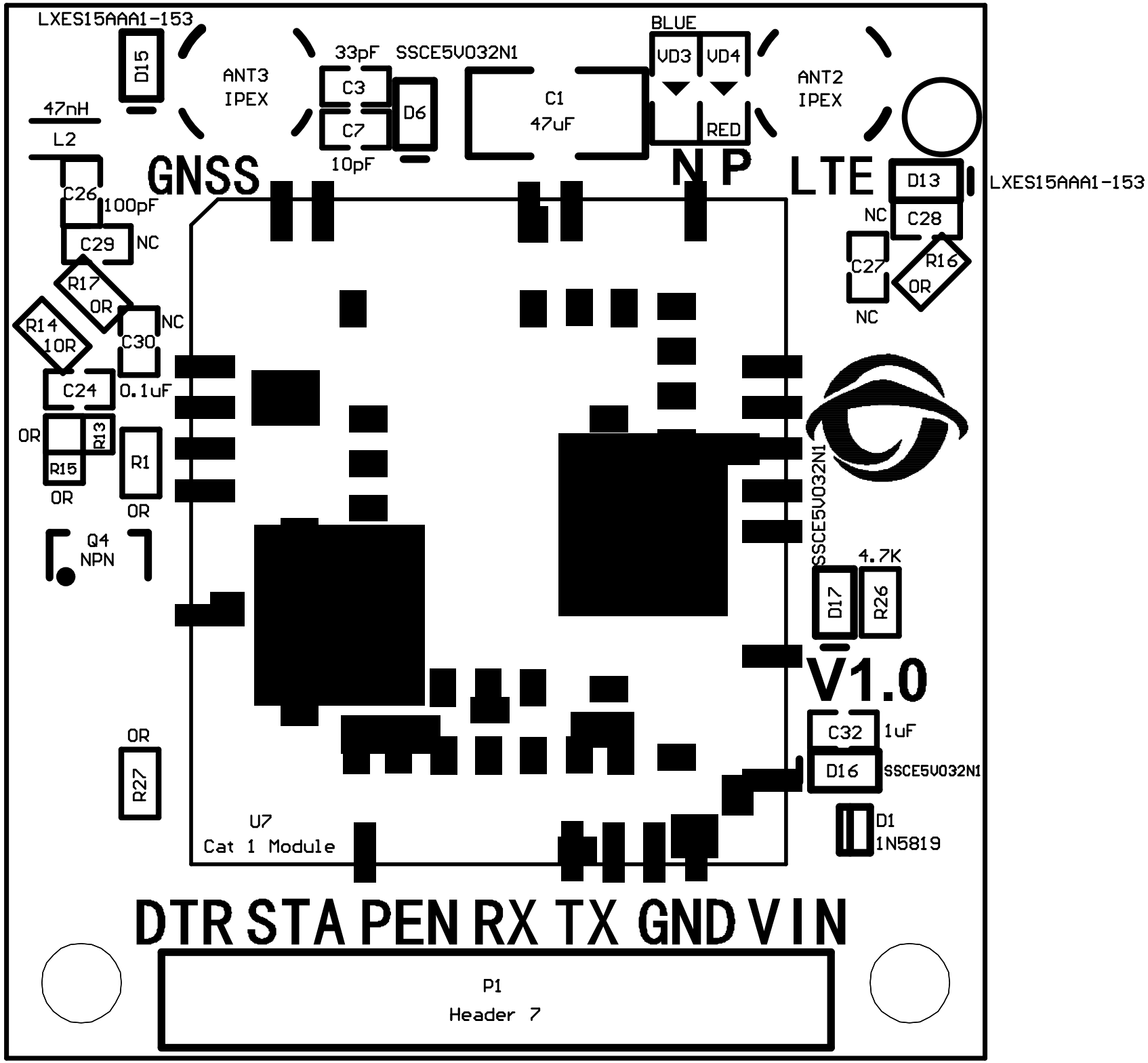

Freestrong
VEXT
BOOT
GND
DN
DP
VBUS
TP7
TP10
TP2
TP6
TP1
TP4
VD2
P
VD1
N
Q3
NPN
RED
R5
1K
R4
1K
BLUE
TP9
VBAT
R30
R31
4.7K
47K
U5
eSIM 5X6
C2
47uF
C5
1uF
U6
LDO_3V3
VCC_3V3
TP3
0.1uF
C6
C16
10uF
C17
10uF
4.7uH
L1
RT5
0R
OR
R25
C15
NS
R35
53.6K 1%
R36
10K 1%
C12
0.1uF
SM8102ABC
U1
RT4
C14
OR 0.1uF
22R
R9
22P
C9
SRV05-4
U3
0.1uF
C8
OR
R24
C10
22P
R8
22R
R12
R18
4.7K
22P
C11
22R
R10
C20
1nF
10K
OR
R20
R28
4.7K
Q6
NPN
Q5
NMOS
7.34K
OR
4.7K
R2
100K
R34
OR
R32
10K
R22
Q2
NPN
D2
1nF
4.7K
C21
R11
47K
R29
R33
OR
OR
R19
D3
D4
R3
4.7K
R23
47K/4.7K
R21
4.7K
R6
R7
Q1
NMOS
C13
22uF
VIN G TX RX EN ST DT
D5
SSCE5V032N1
SSCE5V032N1
SSCE5V032N1
SSCE5V032N1


LXES15AAA1-153
D15
ANT3
IPEX
33pF
SSCE5V032N1
C3
C7
10pF
D6
C1
47uF
BLUE
VD3
VD4
RED
ANT2
IPEX
47nH
L2
GNSS
N P
LTE
D13
LXES15AAA1-153
C26
100pF
C29
NC
NC
C28
C27
NC
R16
OR
R17
OR
R14
10R
C30
NC
C24
0.1uF
OR
R13
R15
OR
R1
OR
Q4
NPN
SSCE5V032N1
4.7K
D17
R26
V1.0
C32
1uF
D16
SSCE5V032N1
D1
1N5819
OR
R27
U7
Cat 1 Module
DTR STA PEN RX TX GND VIN
P1
Header 7